SUPPLY 12V -> 5V
- 5V 5A output
- L, D and polarized C as near as possible to pin PH and GND_2
- Large traces, zones for better performances
- ok traces in input, output just zones in order to avoid drops

SUPPLY 5V -> 3V3

Raspberry

Sensors

File: sensors.kicad_sch

MCU

File: mcu.kicad_sch

Power terminals

Just for current reading

Rotative Encoders

Endstops

Heatbed

IPT-4 DC Motor Drivers

Stepper motors enable

HEADERS

Stepper motors drivers

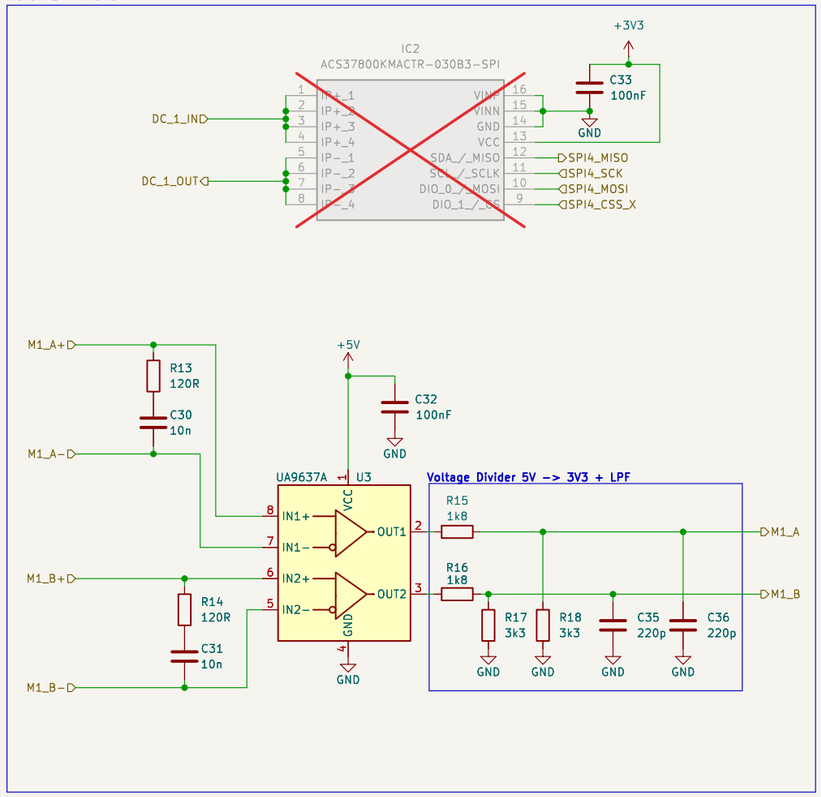
Stepper motors Connectors

Fans

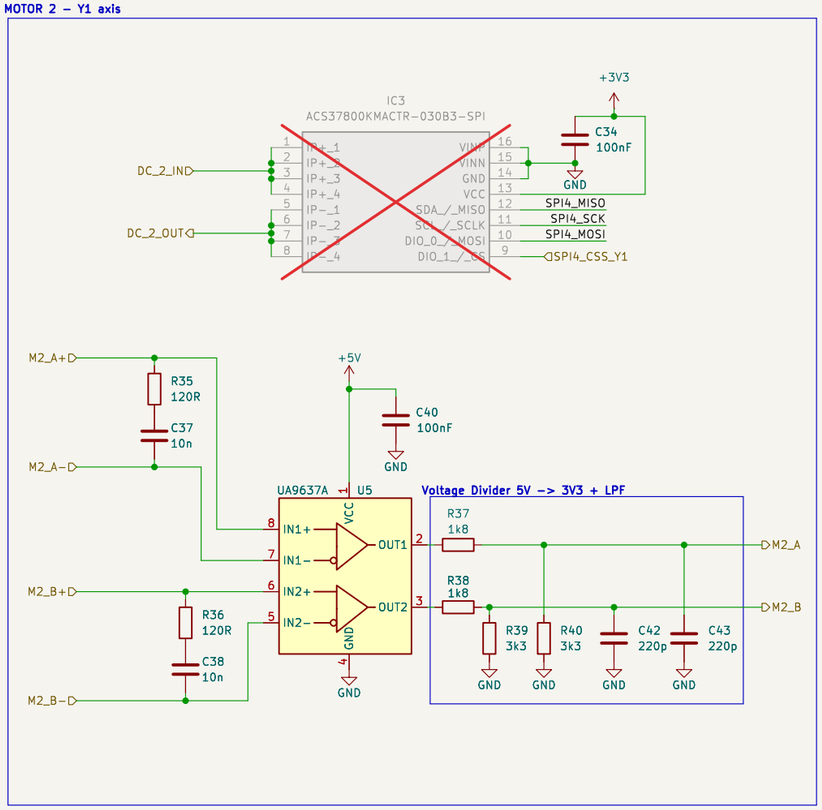
Mounting holes

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Copyright 2026 Leonardo Pasquato	
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Sheet: /	
File: axes-controller-board.kicad_sch	
Title: Dedalus Motherboard	
Size: A2	Date: **Rev: 1.0.0**
KiCad E.D.A. 9.0.7	Id: 1/3

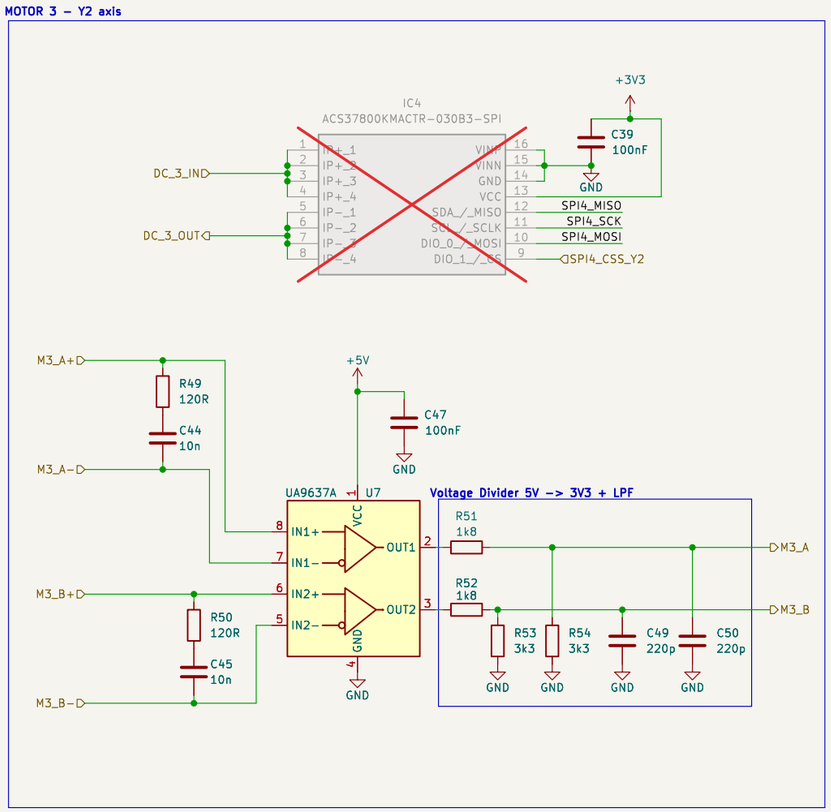
MOTOR 1 – X axis



MOTOR 2 – Y1 axis



MOTOR 3 – Y2 axis



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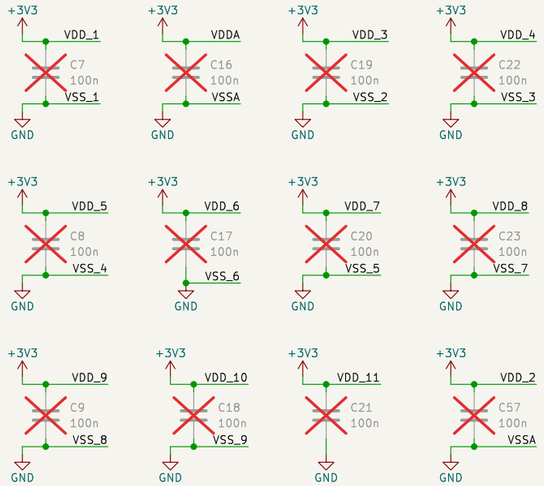
Sheet: /Sensors/
File: sensors.kicad_sch

Title:

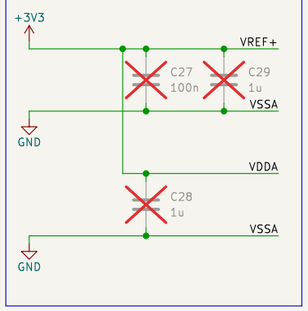
Size: A3	Date:	**Rev: 1.0.0**
KiCad E.D.A. 9.0.7		Id: 2/3

Decoupling

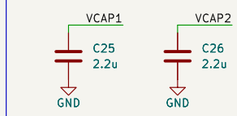
Supply decoupling



Analog supply



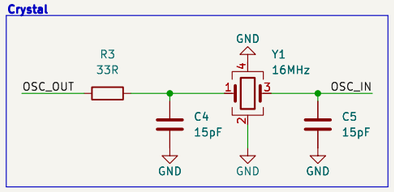
Internal regulator capacitors



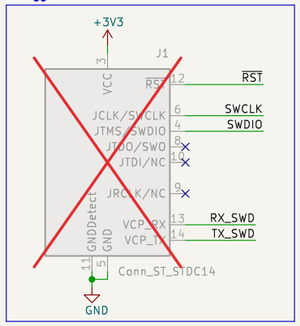
Bulk cap

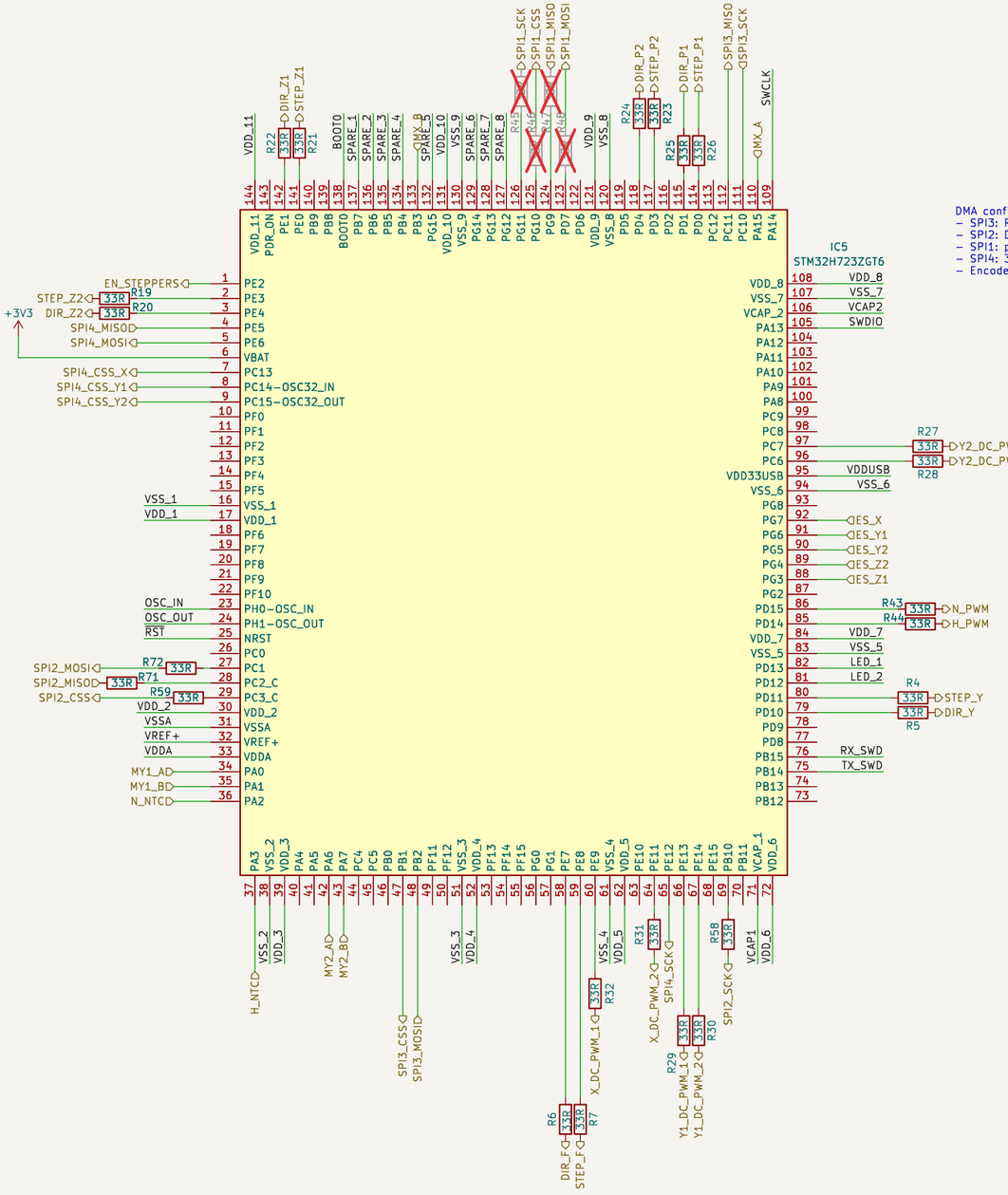
USB

Crystal



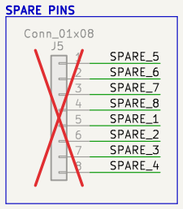
Debugger



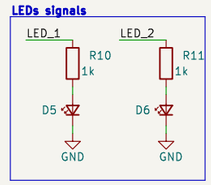


DMA config:
- SPI3: Raspberry communication
- SPI2: DC motors rotative encoders (daisy chain)
- SPI1: pitch and yaw rotative encoders (daisy chain)
- SPI4: 3 current sensors (with CSs)
- Encoders mode for linear encoders

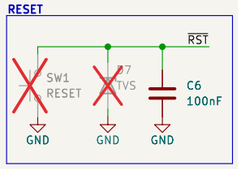
SPARE PINS



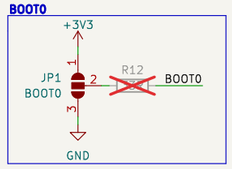
LEDs signals



RESET



BOOT0



PB8-BOOT0 pin configuration:
LOW: normal, boot from user flash
HIGH: boot from system memory, bootloader

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Sheet: /MCU/
File: mcu.kicad_sch

Title:

Size: A3	Date:	**Rev: 1.0.0**
KiCad E.D.A. 9.0.7		Id: 3/3